

Newton–Puisseux Analysis of the Given Polynomial

Consider the plane curve germ at the origin defined by

$$f(x, y) = x^2 + 4x^3 + 6x^4 - 4x^4y + (-2x - 4x^2 - 2x^3)y^2 + y^4 + x^2y^5.$$

Newton–Puisseux theory studies such a germ by encoding the exponents of the monomials into a Newton polygon, extracting candidate exponents from its compact lower edges, and then iteratively refining a Puiseux expansion for each branch. In the standard theory, a local branch is represented by a parametrization $(T^m, p(T))$, or equivalently by a Puiseux series in fractional powers of x ; equivalent parametrizations differ by a root-of-unity change of parameter. The constructive Newton–Puisseux procedure uses the lower Newton polygon, its edge slopes, and the multiplicities of the corresponding edge roots to decide whether a branch splits or whether the computation must continue. ¹

What the algorithm actually uses

For a Newton-convenient polynomial in y , the Newton polygon is the lower convex hull of the support points. In the Canino–Gimigliano–Idà convention, the support is drawn with the x -exponent on one axis and the y -exponent on the other; every compact edge has negative slope, and the candidate exponent for the next Puiseux term is read from that slope. Because different authors swap the axes, the formula for the exponent is sometimes written as $\gamma = -\rho$ and sometimes as $\gamma = -1/\rho$; both are the same statement in different coordinate conventions. The algorithm then substitutes the corresponding trial term, computes the edge polynomial, and checks whether its roots are simple or multiple. Multiple roots force another Newton step; simple roots separate branches immediately. ²

That framework also explains the standard “stop criterion.” In the constructive version given by Canino, Gimigliano, and Idà, each descending path eventually reaches a step of height 1, and at that point the root no longer splits: from then on there is a unique next choice, so the truncation already distinguishes that branch from the others. ³

The correct Newton polygon for this specific polynomial

Write f as a polynomial in y :

$$f(x, y) = a_0(x) + a_1(x)y + a_2(x)y^2 + a_4(x)y^4 + a_5(x)y^5,$$

with

$$a_0 = x^2 + 4x^3 + 6x^4, \quad a_1 = -4x^4, \quad a_2 = -2x - 4x^2 - 2x^3, \quad a_4 = 1, \quad a_5 = x^2.$$

Using the user’s coordinate convention (n, α_n) , where n is the exponent of y and α_n is the smallest exponent of x appearing in $a_n(x)$, the relevant points are obtained by direct inspection:

y -degree n	lowest x -power α_n	point
0	2	(0, 2)
1	4	(1, 4)
2	1	(2, 1)
4	0	(4, 0)
5	2	(5, 2)

The compact lower Newton polygon is **not** naturally read as two different edges $(0, 2) \rightarrow (2, 1)$ and $(2, 1) \rightarrow (4, 0)$. Those three points are collinear, so the lower hull has one compact edge from $(0, 2)$ to $(4, 0)$, with $(2, 1)$ lying on that same edge. The point $(1, 4)$ lies strictly above the hull, and so does $(5, 2)$. By direct calculation, the first relevant slope is therefore

$$\rho = \frac{0 - 2}{4 - 0} = -\frac{1}{2}, \quad \gamma = -\rho = \frac{1}{2}$$

in the user's (n, α_n) convention.

This immediately corrects the first technical issue in the draft derivation: **all monomials lying on the edge must be kept**, not only x^2 and $-2xy^2$. The edge terms are

$$x^2 - 2xy^2 + y^4 = (y^2 - x)^2.$$

So if one substitutes

$$y = Cx^{1/2},$$

the edge equation is

$$x^2 - 2x(Cx^{1/2})^2 + (Cx^{1/2})^4 = x^2(1 - 2C^2 + C^4) = x^2(C^2 - 1)^2.$$

Hence the correct characteristic equation is

$$(C^2 - 1)^2 = 0,$$

not $1 - 2C^2 = 0$. The candidate leading coefficients are therefore $C = \pm 1$, and **each is a double root**. That multiplicity is exactly the signal that the branch has not yet been untangled, so the Newton–Puiseux process must continue rather than stop after the first step. This is precisely the role played by multiple edge roots in the constructive algorithm. ⁴

Carrying out the Newton–Puiseux steps for the example

A convenient way to continue is to put

$$x = t^2, \quad y = tu.$$

Then a direct expansion gives

$$f(t^2, tu) = t^4 \left[(u^2 - 1)^2 - 4t^2(u^2 - 1) + t^4(6 - 2u^2) + t^5(u^5 - 4u) \right].$$

To follow the root $C = 1$, write $u = 1 + v$. Expanding again, the lowest-order terms in (t, v) are

$$4v^2 - 8t^2v + 4t^4 + \text{higher terms} = 4(v - t^2)^2 + \dots$$

so the next Newton step gives

$$v \sim t^2.$$

Thus

$$u = 1 + t^2 + \dots, \quad y = t + t^3 + \dots = x^{1/2} + x^{3/2} + \dots.$$

At this point one sees a second repeated root, so there is still no stopping. Continue by setting $x = S^4$ and

$$y = S^2 + S^6 + z.$$

Direct computation shows that the new lower support points are

$$(0, 18), (1, 16), (2, 4), (3, 2), (4, 0), (5, 8),$$

so the first compact lower edge is now $(0, 18) \rightarrow (2, 4)$. Its slope is -7 , which means the next exponent is 7 in the variable S , i.e. $7/4$ in the original variable x . The corresponding edge polynomial is obtained from the terms on that edge:

$$-3S^{18} + 4S^4z^2,$$

so with $z = CS^7$ one gets

$$-3 + 4C^2 = 0.$$

Therefore

$$C = \pm \frac{\sqrt{3}}{2},$$

and the branch expansion refines to

$$y = x^{1/2} + x^{3/2} \pm \frac{\sqrt{3}}{2}x^{7/4} + \dots.$$

Now choose one sign, say $+\sqrt{3}/2$. A further Newton step gives the next branch-distinguishing term

$$y = x^{1/2} + x^{3/2} + \frac{\sqrt{3}}{2}x^{7/4} + \frac{1}{\sqrt{3}}x^{9/4} + \dots.$$

What matters conceptually is not only the coefficient $\frac{1}{\sqrt{3}}$, but the fact that after the $x^{7/4}$ term the next relevant Newton edge has height 1. In the standard constructive algorithm, that is exactly the situation where the stop criterion applies: the root no longer splits, and the truncation already identifies the branch uniquely among the others. ³

What singularity this actually is

The origin here is **not** a triple point. The lowest total-degree part of f is simply

$$f_2(x, y) = x^2,$$

so the point has multiplicity 2, and its tangent cone is the double line $x = 0$. In the standard branch language, a tangent cone is the product of the tangent lines of the local branches, counted with multiplicity. That already tells you this example belongs to the “double tangent / multiplicity-two” world, not to the “triple point” world discussed later in the user’s summary. ⁵

A reduced local parametrization consistent with the computation above is

$$x = T^4, \quad y = T^2 + T^6 + \frac{\sqrt{3}}{2}T^7 + \frac{1}{\sqrt{3}}T^9 + \dots$$

This parametrization is reduced because the exponents 4, 2, 6, 7, 9, ... have greatest common divisor 1. Its branch multiplicity is

$$m = \min\{\text{ord}(x(T)), \text{ord}(y(T))\} = \min\{4, 2\} = 2,$$

so this is a **2-branch** with tangent line $x = 0$. That fits the tangent-cone computation exactly. ⁶

It is also important not to overcount the Puiseux conjugates. For a branch parametrized by $(T^m, p(T))$, parameter changes $T \mapsto \omega T$ with $\omega^m = 1$ produce equivalent parametrizations. Thus the different Puiseux conjugates coming from $T \mapsto -T, \pm iT$ are not separate analytic branches of the germ; they are the same branch seen through different fourth roots of x . That equivalence is part of the standard branch formalism. ⁷

How the triple-point discussion fits and where it does not

The user’s broader summary of triple points is substantially correct as a description of the **general** output of Newton–Puiseux for a reduced plane curve with a triple point and a triple tangent. In that setting, the algorithm can end with one 3-branch, with one 2-branch plus one 1-branch, or with three 1-branches; repeated edge roots force the analysis to continue, while simple roots separate branches immediately. That classification is stated explicitly in Theorem 4.3 of Canino, Gimigliano, and Idà. ⁸

What does **not** carry over to the present polynomial is the assumption that we are dealing with a triple point. This example is of multiplicity 2, not 3, and its first Newton edge produces the repeated characteristic polynomial $(C^2 - 1)^2$, not the cubic patterns that govern the triple-point theorem. So the triple-point

classification is a good general reference point, but it is not the correct local model for this particular $f(x, y)$. ⁹

Bottom line

For this polynomial, the Newton–Puisseux method does exactly what it is supposed to do, but the draft derivation needs three corrections.

First, the lower Newton polygon has **one** initial compact edge, not two genuinely different ones, because $(0, 2)$, $(2, 1)$, and $(4, 0)$ are collinear. Second, the edge equation must include **all** terms on that edge, so the correct first characteristic equation is

$$(C^2 - 1)^2 = 0,$$

not $1 - 2C^2 = 0$. Third, the origin is **not a triple point**; it is a multiplicity-two singular point with tangent cone x^2 , and the Newton–Puisseux calculation yields a 2-branch with a Puiseux expansion beginning

$$y = x^{1/2} + x^{3/2} + \frac{\sqrt{3}}{2}x^{7/4} + \frac{1}{\sqrt{3}}x^{9/4} + \dots$$

up to choice of Puiseux conjugate. The repeated roots at the first two stages are exactly why the computation must continue, and the later height-1 step is exactly why the constructive stop criterion eventually becomes valid. ¹⁰

¹ ² ⁴ ⁵ ⁶ ⁷ ⁸ [arxiv.org](https://arxiv.org/pdf/2204.05691)

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³ ⁹ ¹⁰ [The Newton–Puisseux Algorithm and Triple Points for Plane Curves | MDPI](https://www.mdpi.com/2227-7390/11/10/2324)

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